

High-Speed Links

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CONTENTS

Contents	i
1 Introduction	1
1.1 Overview	1
1.2	1
2 Channels	5
2.1 Overview	5
2.2	5
3 TDR and S-Parameters	7
3.1 Overview	7
4 Channel Pulse Response and Modeling	11
4.1 Overview	11
4.2 Channel LTI Model	11
5 Termination and Transmitters	13
5.1 Overview	13
6 Receiver Circuits	17
6.1 Overview	17

7	Equalization	21
7.1	Overview	21
7.2	Rx CTLE Equalization	21
8	Adaptive Equalization	25
8.1	Overview	25
9	Noise	29
9.1	Overview	29
10	Jitter	33
10.1	Overview	33
11	Clocking Architectures and PLLs	37
11.1	Overview	37
12	Clock and Data Recovery	41
12.1	Overview	41
13	Forwarded Clocking and Deskew	45
13.1	Overview	45
14	Clock Distribution	49
14.1	Overview	49
15	Die-to-Die Transceivers	53
15.1	Overview	53
16	Optical Transceivers	57
16.1	Overview	57

CHAPTER I

INTRODUCTION

I.1 OVERVIEW

I.2

[Palermo, 2025].

BIBLIOGRAPHY

Sam Palermo. ECEN 720: High-Speed Links Circuits and Systems, 2025. Course lecture notes and slides.

CHAPTER 2

CHANNELS

2.1 OVERVIEW

This chapter introduces the electrical behavior of high-speed channels and the mechanisms that limit signal integrity

2.2

CHAPTER 3

TDR AND S-PARAMETERS

3.1 OVERVIEW

This chapter introduces time-domain reflectometry and S-parameters as tools for characterizing high-speed channels [Palermo, 2025].

BIBLIOGRAPHY

Sam Palermo. ECEN 720: High-Speed Links Circuits and Systems, 2025. Course lecture notes and slides.

CHAPTER 4

CHANNEL PULSE RESPONSE AND MODELING

4.1 OVERVIEW

This chapter develops time-domain channel models and examines the impulse and pulse responses of communication channels

4.2 CHANNEL LTI MODEL

Assume input discrete-time sequence is $x[n]$, corresponding continuous-time signal $x(t)$ could be written as

$$x(t) = \sum_{n=-\infty}^{\infty} x[n] w_T(t - nT) \quad (4.1)$$

where $w_T(t)$ is a window function with unit amplitude and duration T , corresponding to one symbol period.

$$w_T(t) = u(t) - u(t - T). \quad (4.2)$$

The received signal is

$$y(t) = x(t) * h_{ch}(t) = \sum_{n=-\infty}^{\infty} x[n] w_T(t - nT) * h_{ch}(t) \quad (4.3)$$

Define the Channel Pulse Response as

$$p_{ch}(t) = w_T(t) * h_{ch}(t). \quad (4.4)$$

Then

$$y(t) = \sum_{n=-\infty}^{\infty} x[n] p_{ch}(t - nT). \quad (4.5)$$

Let t_o be the time at which $p_{ch}(t)$ reaches its peak, and define

$$p_{ch}[n] = p_{ch}(t_o + nT). \quad (4.6)$$

Thus,

$$p_{ch}[0] = p_{ch}(t_o) \quad (4.7)$$

is the main cursor.

Sampling the received signal at

$$t = t_o + kT, \quad (4.8)$$

gives

$$y[k] = y(t_o + kT). \quad (4.9)$$

Therefore,

$$y[k] = \sum_{n=-\infty}^{\infty} x[n] p_{ch}(t_o + (k - n)T). \quad (4.10)$$

Using the definition of $p_{ch}[n]$,

$$y[k] = \sum_{n=-\infty}^{\infty} x[n] p_{ch}[k - n]. \quad (4.11)$$

Hence,

$$y[k] = x[n] * p_{ch}[n]. \quad (4.12)$$

CHAPTER 5

TERMINATION AND TRANSMITTERS

5.1 OVERVIEW

This chapter introduces transmission-line termination techniques and the circuit principles of high-speed transmitters [Palermo, 2025].

BIBLIOGRAPHY

Sam Palermo. ECEN 720: High-Speed Links Circuits and Systems, 2025. Course lecture notes and slides.

CHAPTER 6

RECEIVER CIRCUITS

6.1 OVERVIEW

This chapter introduces the principal circuit structures used to detect and recover high-speed signals at the receiver [Palermo, 2025].

BIBLIOGRAPHY

Sam Palermo. ECEN 720: High-Speed Links Circuits and Systems, 2025. Course lecture notes and slides.

CHAPTER 7

EQUALIZATION

7.1 OVERVIEW

This chapter introduces equalization techniques used to compensate for channel loss and inter-symbol interference

7.2 Rx CTLE EQUALIZATION

Start from a typical CTLE response,

$$H_{\text{CTLE}}(s) = A_o \frac{1 + s/\omega_z}{1 + s/\omega_p}, \quad \omega_z < \omega_p \quad (7.1)$$

$$H_{\text{CTLE}}(s) = A_o \frac{\omega_p}{\omega_z} \left[1 - \frac{\omega_p - \omega_z}{s + \omega_p} \right] \quad (7.2)$$

The first term, 1, represents the signal itself, while the second term, $1/(s + \omega_p)$, represents a low-pass filter.

Therefore, the CTLE can be interpreted as the original signal minus its low-pass-filtered component.

Now we take a look at how CTLE cancel post-cursor in time domain. First write the impulse response of CTLE, which is inverse-transform of transfer function $H_{\text{CTLE}}(s)$

$$h_{\text{CTLE}}(t) = K\delta(t) - Ce^{-\omega_p t}u(t) \quad (7.3)$$

where

$$K = A_o \frac{\omega_p}{\omega_z}, \quad C = A_o \frac{\omega_p}{\omega_z} (\omega_p - \omega_z). \quad (7.4)$$

For channel pulse response $p_{\text{ch}}(t)$,

$$p_{\text{eq}}(t) = p_{\text{ch}}(t) * h_{\text{CTLE}}(t) \quad (7.5)$$

thus

$$p_{\text{eq}}(t) = Kp_{\text{ch}}(t) - C \int_{-\infty}^t p_{\text{ch}}(\tau) e^{-\omega_p(t-\tau)} d\tau. \quad (7.6)$$

Define

$$m(t) = \int_{-\infty}^t p_{ch}(\tau) e^{-\omega_p(t-\tau)} d\tau, \quad (7.7)$$

then

$$p_{eq}(t) = Kp_{ch}(t) - Cm(t). \quad (7.8)$$

Since $m(t)$ is a low-pass filtered version of $p_{ch}(t)$, the fast-changing main cursor is less cancelled, while the slowly varying post-cursor tail is more strongly cancelled.

From Chapter 2

$$p_{ch}[n] = p_{ch}(t_o + nT). \quad (7.9)$$

Define

$$p_{eq}[n] = p_{eq}(t_o + nT). \quad (7.10)$$

Since

$$p_{eq}(t) = Kp_{ch}(t) - Cm(t), \quad (7.11)$$

we have

$$p_{eq}[n] = Kp_{ch}[n] - Cm(t_o + nT), \quad (7.12)$$

and

$$p_{eq}[0] = Kp_{ch}[0] - Cm(t_o). \quad (7.13)$$

To show

$$\frac{p_{eq}[n]}{p_{eq}[0]} < \frac{p_{ch}[n]}{p_{ch}[0]}, \quad (7.14)$$

substitute the above expressions:

$$\frac{Kp_{ch}[n] - Cm(t_o + nT)}{Kp_{ch}[0] - Cm(t_o)} < \frac{p_{ch}[n]}{p_{ch}[0]}. \quad (7.15)$$

After cross multiplication and cancellation,

$$p_{ch}[0] m(t_o + nT) > p_{ch}[n] m(t_o). \quad (7.16)$$

Equivalently,

$$\frac{m(t_o + nT)}{p_{ch}[n]} > \frac{m(t_o)}{p_{ch}[0]}. \quad (7.17)$$

Since $m(t)$ is a low-pass filtered version of $p_{ch}(t)$, it follows the slowly varying post-cursor more strongly than the fast-changing main cursor. Therefore,

$$\frac{p_{\text{eq}}[n]}{p_{\text{eq}}[0]} < \frac{p_{\text{ch}}[n]}{p_{\text{ch}}[0]}, \quad n = 1, 2, 3, \dots \quad (7.18)$$

CHAPTER 8

ADAPTIVE EQUALIZATION

8.1 OVERVIEW

This chapter introduces adaptive techniques for automatically adjusting receiver equalization to changing channel conditions [Palermo, 2025].

BIBLIOGRAPHY

Sam Palermo. ECEN 720: High-Speed Links Circuits and Systems, 2025. Course lecture notes and slides.

CHAPTER 9

NOISE

9.1 OVERVIEW

This chapter examines the principal noise sources that limit the performance of high-speed communication links [Palermo, 2025].

BIBLIOGRAPHY

Sam Palermo. ECEN 720: High-Speed Links Circuits and Systems, 2025. Course lecture notes and slides.

CHAPTER 10

JITTER

10.1 OVERVIEW

This chapter introduces timing jitter, its principal sources, and its effects on high-speed link performance [Palermo, 2025].

BIBLIOGRAPHY

Sam Palermo. ECEN 720: High-Speed Links Circuits and Systems, 2025. Course lecture notes and slides.

CHAPTER II

CLOCKING ARCHITECTURES AND PLLs

II.1 OVERVIEW

This chapter introduces clocking architectures and phase-locked loops used in high-speed communication systems [Palermo, 2025].

BIBLIOGRAPHY

Sam Palermo. ECEN 720: High-Speed Links Circuits and Systems, 2025. Course lecture notes and slides.

CHAPTER 12

CLOCK AND DATA RECOVERY

12.1 OVERVIEW

This chapter introduces clock and data recovery techniques for extracting sampling timing from received data [Palermo, 2025].

BIBLIOGRAPHY

Sam Palermo. ECEN 720: High-Speed Links Circuits and Systems, 2025. Course lecture notes and slides.

CHAPTER 13

FORWARDED CLOCKING AND DESKEW

13.1 OVERVIEW

This chapter examines forwarded-clock architectures and deskew techniques for maintaining timing alignment between signals [Palermo, 2025].

BIBLIOGRAPHY

Sam Palermo. ECEN 720: High-Speed Links Circuits and Systems, 2025. Course lecture notes and slides.

CHAPTER 14

CLOCK DISTRIBUTION

14.1 OVERVIEW

This chapter introduces the distribution of high-speed clocks and the timing limitations associated with clock networks [Palermo, 2025].

BIBLIOGRAPHY

Sam Palermo. ECEN 720: High-Speed Links Circuits and Systems, 2025. Course lecture notes and slides.

CHAPTER 15

DIE-TO-DIE TRANSCEIVERS

15.1 OVERVIEW

This chapter introduces high-speed die-to-die communication and the transceiver architectures used in advanced packages [Palermo, 2025].

BIBLIOGRAPHY

Sam Palermo. ECEN 720: High-Speed Links Circuits and Systems, 2025. Course lecture notes and slides.

CHAPTER 16

OPTICAL TRANSCEIVERS

16.1 OVERVIEW

This chapter introduces optical communication links and the electrical circuits used to interface electronic and optical signals [Palermo, 2025].

BIBLIOGRAPHY

Sam Palermo. ECEN 720: High-Speed Links Circuits and Systems, 2025. Course lecture notes and slides.